PEWA-F02-R01
Comprehensive Food & Beverage Menu Classification Research

Independent Research Consultant Report — Foundation Stage F02

Status: Research only; no implementation authorised

1. Executive Summary

This report presents evidence-based research into how food and beverage menus are classified, organised, presented and searched across global and African food-service businesses. The objective is a proposed standard menu classification library for PEWA that remains customisable by each participating business.

Key findings: (1) Global practice converges on 5–7 course-oriented categories (Appetizers, Soups, Salads, Mains, Sides, Desserts, Beverages) with protein or cuisine sub-groupings. (2) East African (Kenya, Tanzania, Uganda, Rwanda, Burundi) menus are staple-led rather than course-led; culturally meaningful categories such as Staples/Swallow, Stews & Sauces, Leafy Greens, Bean Dishes and Nyama Choma must be preserved as first-class. (3) Multi-location operators use a single item master with controlled location, price and availability overrides. (4) Allergen and dietary attributes require strict factual governance distinct from marketing claims. (5) No single global standard taxonomy exists; PEWA should offer a Core library plus Regional and Optional layers.

2. Methodology

Sources included: published restaurant and hotel menu guidance (UpMenu, industry textbooks); ISO 16520:2025 restaurants and catering vocabulary; East African culinary and travel-food references (2023–2026) covering Kenya, Tanzania, Uganda, Rwanda and Burundi; West African (Nigeria, Ghana), Southern African (South Africa, Zimbabwe) and North African (Morocco, Maghreb) dish and structure descriptions; café, bakery and beverage menu practice guides; multi-location POS and menu-management literature; allergen regulatory frameworks (EU 14, US 9+sesame, selected African references). No primary field observation of physical menus was performed; evidence is secondary and triangulated where possible.

Distinction of evidence types used throughout:

- Directly observed evidence — statements taken from cited sources describing actual menus or regulations.
- Common industry practice — patterns repeated across multiple independent sources.
- Consultant interpretation — synthesis or recommendation not explicitly stated in a single source.
- Proposed PEWA recommendation — design suggestion for the platform library.

3. Geographical and Business-Type Coverage

Geography: Global (Western/international patterns); East Africa (strong primary focus — Burundi, Rwanda, Uganda, Kenya, Tanzania); Central, West, Southern and North Africa (secondary). Business types: casual dining, fine dining, quick-service, hotels, cafés, bakeries/patisseries, bars, food halls, takeaways, delivery-only kitchens, multi-location chains and franchises.

4. Detailed Findings

A. Global Food Menu Classification

Observed common categories (industry practice, multiple sources): Appetizers/Starters, Soups, Salads, Main Courses/Entrées, Side Dishes, Desserts, Beverages. Additional frequent sections: Sandwiches/Burgers, Pizza, Pasta, Specials, Children's, Combos. Service-style guidance typically recommends 4–7 core categories with limited items per category to avoid choice overload. Fine-dining menus often follow classic course progression; QSR and fast-casual emphasise protein or product-type groupings (Burgers, Chicken, Sides, Drinks). ISO 16520:2025 provides vocabulary (à la carte, children's menu, accompaniments, combination menu, ethnic menu, etc.) but does not prescribe a hierarchical taxonomy of dish categories.

Source note: UpMenu category guides; ISO 16520:2025; restaurant management textbooks.

B. African and Regional Food Classification

East Africa (high evidence): Meals are structured around a staple (ugali/ubugali/posho in Kenya/TZ/RW/BI; matooke in Uganda; variants of maize, cassava, millet or plantain) + stew/sauce + greens or beans + optional protein (nyama choma, fish, chicken). Forcing these into Western 'Sides' or 'Mains' loses cultural meaning. Recommended first-class categories: Staples/Swallow, Stews & Sauces, Leafy Greens & Vegetables, Bean & Legume Dishes, Grilled/Nyama Choma, Street Food & Snacks, Rice Dishes (pilau, wali wa nazi). Key dishes observed across sources: ugali, sukuma wiki, nyama choma, matoke/luwombo, isombe, ibiharage/maharagwe, brochettes, samosas/sambusas, mandazi, chapati, Rolex, chipsi mayai, mukeke/sambaza (Burundi lake fish).

West Africa: Soup + swallow (fufu, banku, eba, pounded yam) is structural; jollof rice, egusi, groundnut/peanut stews, suya, red red, kelewele are prominent. Southern Africa: Sadza/pap + relish; braai culture; chakalaka; mopane worms in some contexts. North Africa: Tagine, couscous, harira, pastilla, mezze-style salads; spice and preserved-lemon profiles. Language variations: ugali = ubugali (Kirundi/Kinyarwanda) = posho (Uganda) = sadza (Zimbabwe) = pap/mieliepap (South Africa). Preserve alternative names.

Source note: East African culinary guides 2023–2026; Wikipedia cuisine pages; travel-food atlases; West/Southern/North African dish lists.

C. Beverages and Specialist Food Businesses

Cafés/coffee shops: hierarchical around Espresso-based, Brewed/Filter, Cold brew/Iced, Tea, Other hot, Soft drinks, then Food (pastries, light meals). Modifiers (milk type, syrup, size, temperature) are critical and independent of category. Bakeries: Breads, Pastries/Viennoiserie, Cakes/Gateaux, Cookies/Small bakes, Savoury items, Beverages. Bars: Beer, Wine, Spirits, Cocktails, Soft/Mocktails; food often limited to sharing plates or bar snacks. Juice bars: Juices, Smoothies, Bowls, optionally light food. Overlap with restaurant beverage sections is high; specialist venues simply elevate beverage hierarchy.

D. Menu Structure and Presentation

Recommended conceptual separation (consultant interpretation + industry practice):

- Menu Collection / Named Menu — a published set of items (e.g. 'Lunch Menu', 'Dinner', 'Ramadan Special'). Independent of category hierarchy.
- Meal Occasion / Service Period — Breakfast, Lunch, Dinner, All-day, Late-night. Independent dimension; controls availability.
- Course — classical sequence position (Starter → Main → Dessert). Often mirrored by category order.
- Category / Subcategory — navigational and reporting grouping (hierarchical).
- Menu Section — presentation grouping on a specific menu (may map to categories or be free-form).
- Individual Menu Item — unique sellable product (the item master).
- Combo / Bundle — commercial packaging of multiple items.
- Side / Accompaniment — can be category or linked modifier.
- Add-on / Modifier — option that alters an item (size, extra, cooking preference); not a category.
- Tag / Attribute — searchable, filterable descriptors (independent dimensions).

Hierarchical: Category → Subcategory. Independent dimensions: Occasion, Course (optional), Tags, Modifiers, Price, Availability.

E. Tags, Attributes and Customer Filters

Factual attributes (allergens, presence of specific ingredients when audited) must be governed separately from business-declared claims (Vegetarian, Vegan, Halal, 'Traditional', Spice Level). EU requires 14 allergens; US 9 (+ sesame); African markets vary and often lack restaurant-specific mandates comparable to EU. PEWA should default to a Core allergen set (the common 9 + strong emphasis on peanuts/sesame given African cuisine prevalence) and allow market-specific expansion. Multilingual synonyms and regional naming (ugali/posho/sadza; groundnuts/peanuts) are essential for search. Safeguard: never auto-generate allergen tags from free-text descriptions.

F. Multi-Location and Reusable Menu Management

Industry practice (multiple POS/menu-management sources): maintain one item master; assign items to menus, locations and channels; allow controlled overrides for price, availability, local specials and taxes. Full duplication of items per location is the primary source of drift and reporting failure. Versioning and scheduled publish (seasonal, day-part) are common. Franchise models often lock core recipes while permitting local price or limited specials.

G. Business Customisation and Taxonomy Governance

Proposed principles: (1) Core library is default and shared; (2) businesses may add custom categories and map them to Core IDs for analytics; (3) renaming for local language is encouraged; (4) retirement of Core entries only after review cycle; (5) avoid uncontrolled synonym proliferation by maintaining a governed synonym table; (6) PEWA does not force every business to use the full library.

H. Customer Discovery and Search

Search targets three entities: businesses, menus, and individual items. Useful signals: cuisine/region tags, dietary/allergen filters, meal occasion, price band, availability, spice, preparation method, free-text with synonym expansion. Regional cuisine filters are especially valuable in multi-ethnic African cities.

5. Recommendations for PEWA Classification Library

1. Implement a Core hierarchical category set (Course Structure + East African Structure + Beverage + Bakery families) with Regional extensions (West, Southern, North African).
2. Keep meal occasions, dietary/allergen tags, cuisine, spice, portion and modifiers as independent dimensions.
3. Enforce a single item master; menus and locations only reference items.
4. Treat allergen data as factual and audited; separate from marketing claims.
5. Support business-specific categories with optional mapping to Core IDs.
6. Maintain synonym and alternative-name tables for search (especially East African staples).

6. Gaps and Unresolved Questions

Limited primary menu examples from Burundi and rural Central Africa; sparse formal restaurant-taxonomy standards in most African markets; quantitative customer search behaviour by region not located; North/West/Southern African menu section structures less documented than dish lists. Open PEWA decisions: mandatory allergen set for multi-country tenants; depth of Core vs Optional; modelling of staple+sauce as one configurable item vs two linked items; governance process for local terminology additions.

— End of Research Report —

Files cross-referenced: Classification Catalogue XLSX; Tags & Filters Catalogue XLSX; Menu Structure Examples PDF; Reconciliation & Gap Report PDF.